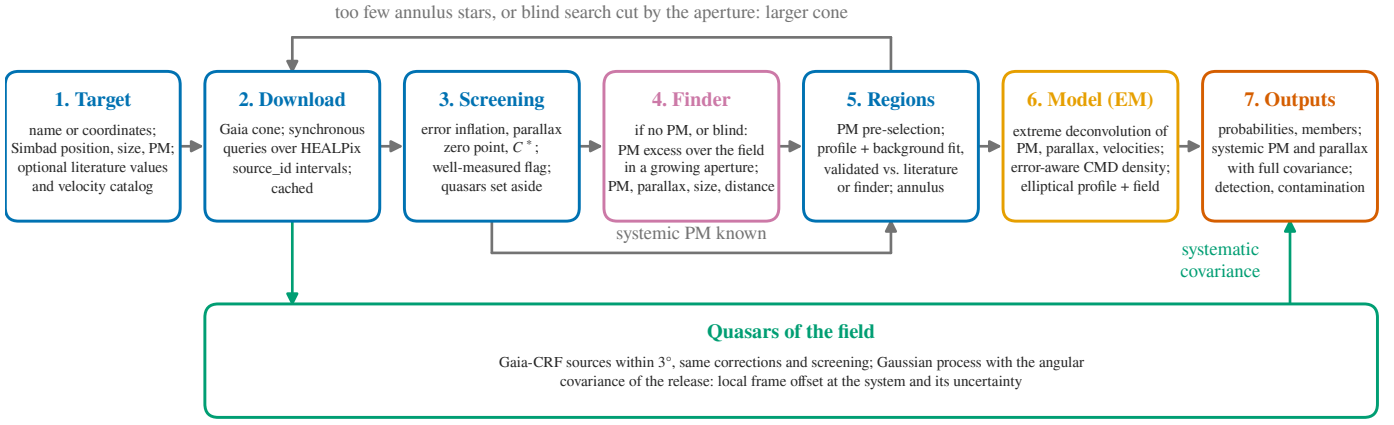
too few annulus stars, or blind search cut by the aperture: larger cone
1. Target
name or coordinates; Simbad position, size, PM; optional literature values and velocity catalog
2. Download
Gaia cone; synchronous queries over HEALPix source_id intervals; cached
3. Screening
error inflation, parallax zero point, C^*; well-measured flag; quasars set aside
4. Finder
if no PM, or blind: PM excess over the field in a growing aperture; PM, parallax, size, distance
5. Regions
PM pre-selection; profile + background fit, validated vs. literature or finder; annulus
6. Model (EM)
extreme deconvolution of PM, parallax, velocities; error-aware CMD density; elliptical profile + field
7. Outputs
probabilities, members; systemic PM and parallax with full covariance; detection, contamination
systemic PM known
systematic covariance
Quasars of the field
Gaia-CRF sources within 3°, same corrections and screening; Gaussian process with the angular covariance of the release: local frame offset at the system and its uncertainty